Airtel Family Safety — System, API & Integration Plan

Purpose: From the BRDs, EA HLD docs and the Figma Exhaustive-Cases flows, identify **every system** that participates in Family Safety / Safety Circle, **which APIs each system exposes/consumes**, **how the systems interact**, the **flow with the Channel Backend (BFF) and the user/party-profile backend (UCM) in focus**, the **engineering effort** (by API + complexity), the **challenges**, and the **partners**.

Status: Draft for Architecture + Product review **Date:** 2026-10-30 **Based on:**

- `Airtel_Family_Safety_BRD_(002)(1).pdf` — product BRD (circle, check-in, SOS, unreachability, routine, fraud, parental, SafeTrip)
- `EA-Create_Family_Safety_Circle.pdf` — HLD: circle setup, consent, membership lifecycle, UCM data model
- `EA-Family_Safety_Architecture.pdf` — HLD: commercial plane + Device/Network Safety (Whalebone, COM, Subscription), **Channel Backend (BFF)**
- `BRD-_SDK_solution_2nd_app_.pdf` — partner SDK / license management BRD
- `EA-Safety_Trip_-_Real-time_GPS_Location.pdf` — Location Hub + Trip Manager
- Figma `39:4390` Exhaustive Cases (≈487 LP screens) + `Family-Safety-Circle-Functionality-Checklist.md`

> **Legend for API maturity** `(EA)` = named in an EA HLD doc · `(BRD)` = stated in a BRD · `[INFERRED]` = required by Figma/BRD behaviour but not yet in a shared LLD. Finalise all paths in OpenAPI during LLD.

---

0. Answer up front — the two backends you asked to focus on

You asked about	What the source docs actually name	Where
Channel Backend	**Channel Backend (BFF)** — `<<channel>>` tier that sits between the Thanks App and the Service Order Manager (COM) / Subscription Manager / Whalebone / Network APIs. It is the single aggregation + auth tier for all Family Safety journeys.	EA-Family_Safety_Architecture — "Enable/Disable Device & Network Safety" sequence diagrams
User-Profile Backend	**Not named verbatim.** The equivalent role is split across **UCM** (new *Family Circle* entity + *Party Master / Individual* records = the member/party profile store) and **Circle Manager** (owns circle + membership lifecycle on top of UCM). Commercial identity still lives in **Siebel/Customer (TMF629)**; family hierarchy comes from **Optimus API**.	EA-Create_Family_Safety_Circle — UCM Family Circle Domain + Party Master diagram

So in this plan, **"user-profile backend" = UCM Party Master + UCM Family Circle entity, fronted by Circle Manager**, and every member-identity read/write is routed there (never into the Customer/TMF629 billing entity — see §9 "Why Customer entity fails").

---

1. System inventory (who is in play)

1.1 Family Safety Circle plane (consent, circle, safety events, location)

ID	System	Owner (from EA stakeholder matrix)	Responsibility
S01	**Thanks App** (+ Safety App for device protection)	App FE / Marketing (Lakshay)	Discovery, purchase, circle UI, SOS widget, dashboard, SDK host
S02	**Channel Backend (BFF)** ⭐	Channel (Surili Jain)	Single aggregation/auth tier: fans out to Circle Manager, Safety Manager, Trip Manager, COM, Subscription, Network; builds home/member-card payloads
S03	**Circle Manager / Family Circle Service**	UCM (Vibhu Gupta)	Circle entity, invites, membership lifecycle, roles, consent, member validation for Trip
S04	**UCM — Party Master + Family Circle entity** ⭐	UCM (Vibhu Gupta)	Party/Individual records (MSISDN, name), new *FamilyCircle* entity holding circle metadata + member Party IDs + roles + per-member feature characteristics
S05	**Safety Manager**	EA (Safety Event model)	Unified Safety Event: `CHECK_IN`, `SOS`, `GEOFENCE`, `INACTIVITY`, `FRAUD`, `FALL`; state machine, responses, history
S06	**Trip Manager**	EA (Safety Trip)	SafeTrip / Emergency Trip create/start/stop, viewer selection, tracking-session link
S07	**Location Hub**	EA (Safety Trip)	Location ingest API, latest-location API, SSE stream; Kafka `location.events`, Redis/Aerospike latest, History DB
S08	**Network API**	Network (Gopal Dave / Bhabani)	Reachability/network state, cell-tower location assist, **Airtel Safe fraud signal feed**
S09	**Device Telemetry**	Device platform	Battery, signal/mode, ringer, unlock/activity, permission state
S10	**CLM**	Lifecycle	Staged escalation journeys (30→45→60 min) from clickstream miss events
S11	**IVR / BumbleBee voice**	Voice	Automated check-in call (60 min); invite "press 1" join
S12	**BumbleBee Communication Platform (Push + SMS)**	Comms	Invites, reminders, SOS, threat alerts, activation msgs
S13	**OTP / Bouncer**	Auth	Web-invite consent OTP; OTP/biometric for role/permission changes (SC7)
S14	**Admin Console**	Ops	Escalation thresholds (CI4), nudge config (EX2)
S15	**Advisor / CRM view**	Service	Read-only case/consent/alert visibility (BRD §11)
S16	**Clickstream / Analytics**	Analytics	Instrumentation (BRD §14); feeds CLM windows
S17	**Audience Manager**	Audience (Parshant Verma)	Free-trial / "Claim Now" eligibility rules

1.2 Commercial / BSS plane (pack purchase, subscription, license)

ID	System	Owner	Responsibility
S18	**COM / Service Order Manager**	COM (Sidhant / Vikalp)	**All LOB orders via TMF622**; orchestrates provisioning
S19	**Subscription Manager (ENDMA)**	Digital (Salil / Kaushal)	Family Safety subscription carrying **two services: Device Safety + Network Safety**
S20	**EPC (Product/Service catalog) + Promotion**	Product (Shubham/Shadab)	Product & Service Specs (`partner_sku_id` , planTier, screens), hidden free-sample PO
S21	**Siebel Postpaid / ATOM / FX / Inform / LAPU / Event Bus / M2M Product Inventory**	BSS (Sai/Nitish/Gaurav/Ankit)	Postpaid+prepaid provisioning rails, customer entity, order ingestion
S22	**Optimus API + Family Mode**	BSS/CX	Parent–child / managed-account hierarchy; add MSISDN to admin Family Mode
S23	**Whalebone Partner backend**	Partner (external)	License allocate/verify/suspend/delete; threat event sync
S24	**Whalebone SDK (on-device)**	Partner (external)	Malware, fraud-APK block, safe browsing, content filter, parental enforcement, Device ID

2. Partners (as requested)

#	Partner	Type	What they provide	APIs / interface	MVP?
1	**Whalebone**	External security vendor (SDK + backend)	Malware detection, fraud-APK block, safe browsing, content filtering, parental enforcement, Device ID; **annual license pool** rotated into Airtel packs	License allocate / map / verify / suspend / delete; threat-event sync to Airtel dashboard; on-device SDK as a **dynamically loaded module inside Thanks App**	✅ Yes (Device/Network Safety)
2	**24×7 Fraud Expert Helpline**	BPO / service desk	Human experts to verify suspicious activity / report fraud	Call-centre integration (telephony + case view)	✅ (SDK BRD Feature 5)
3	**Uber / Ola / Rapido**	Mobility	Trip origin/destination/ETA/route for destination-aware SafeTrip	Partner trip APIs	❌ Phase 2+ (explicitly NON-MVP)
4	**Google Maps**	Maps/geocoding	Route-specific monitoring, detours	Maps/Directions API	❌ Phase 2+ (NON-MVP)
5	**Wearable partner(s)**	Device	Fall / movement / inactivity signals	Partner wearable API	❌ Future (if integrated)

> **BumbleBee** and **IVR** are Airtel-internal platforms (Push/SMS/voice), not external partners. **Whalebone is the only external partner in MVP.**

---

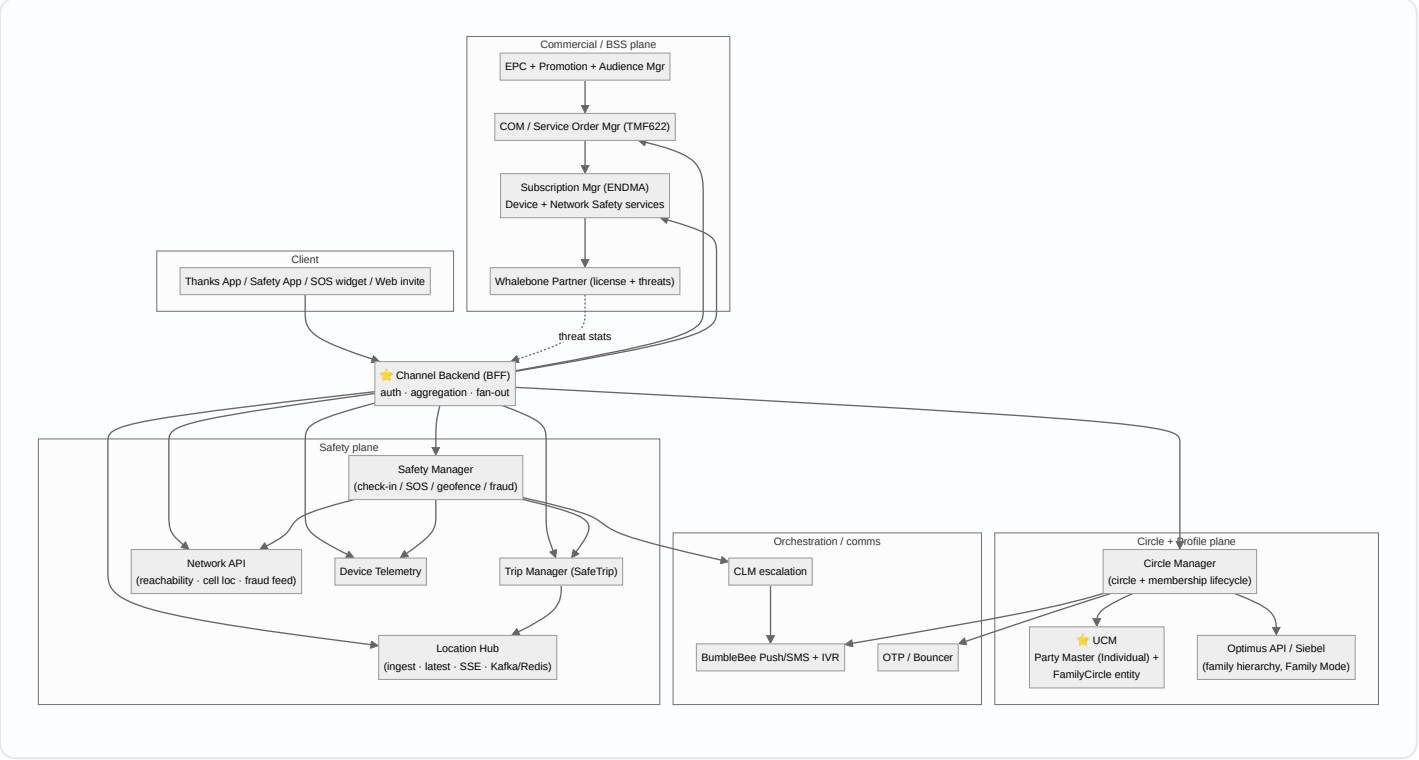
3. System interaction — connection matrix

From	To	Protocol	Why
Thanks App	**Channel Backend (BFF)**	REST / SSE	All journeys go through BFF (auth + aggregation)
Channel Backend (BFF)	Circle Manager	REST	Create circle, invite, accept, leave, roles, members
Channel Backend (BFF)	Safety Manager	REST	Check-in, SOS, responses, history, schedules
Channel Backend (BFF)	Trip Manager	REST	Start/stop SafeTrip, select viewers
Channel Backend (BFF)	Location Hub	REST + **SSE**	Latest location + live stream to viewers
Channel Backend (BFF)	COM (Service Order Mgr)	REST (TMF622)	Buy/claim pack
Channel Backend (BFF)	Subscription Mgr (ENDMA)	REST	Device/Network Safety toggle (config, not order)
Channel Backend (BFF)	Network API / Telemetry	REST	Reachability + device status for member cards
Circle Manager	**UCM Party Master / Family Circle entity**	REST	Persist circle + member Party IDs + roles + feature chars
Circle Manager	TMF632 Individual	REST	Resolve/create Individual (party) by phone number
Circle Manager	Optimus API / Siebel	REST	Fetch family hierarchy; add MSISDN to admin Family Mode
Circle Manager	BumbleBee (Push/SMS) / IVR	Async	Invite dispatch (app / SMS / press-1), member-left notice
Circle Manager	OTP / Bouncer	REST	Web-invite consent OTP; SC7 auth
Circle Manager	Subscription Mgr	Event	Revoke shared pack on member/admin exit
Safety Manager	BumbleBee Push	Async	Check-in / SOS notify
Safety Manager	CLM	Event	Start escalation journey on miss
Safety Manager	Trip Manager → Location Hub	REST	Emergency Trip on SOS
Safety Manager	Network + Telemetry	REST	Context on SOS / unreachability
CLM	Push → sticky → IVR → Safety Manager	Orchestration	30→45→60 min escalation then alert circle
Trip Manager	Circle Manager	REST	Validate viewers are active members
Trip Manager	Location Hub	REST	Create/close tracking session
Location Hub	Kafka → Redis/Aerospike + History DB	Internal	Latest-location fan-out for SSE

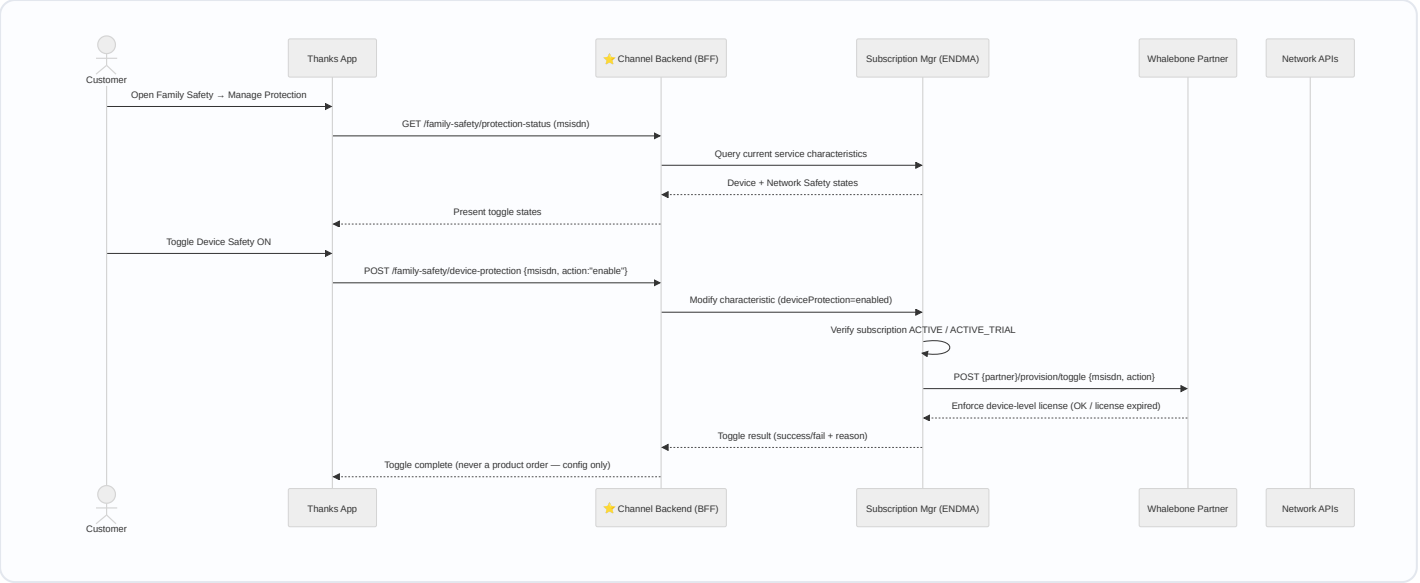
From	To	Protocol	Why
COM	Subscription Mgr → Whalebone	Order events / Partner API	Activate subscription, assign license
Whalebone SDK	Whalebone Partner backend	Partner channel	License verify, threat telemetry
Whalebone Partner	Channel Backend / Dashboard API	Partner → Airtel	Threats-blocked stats sync
IVR / BumbleBee	Safety Manager	Callback	Record IVR response on check-in/invite

4. Flow diagrams

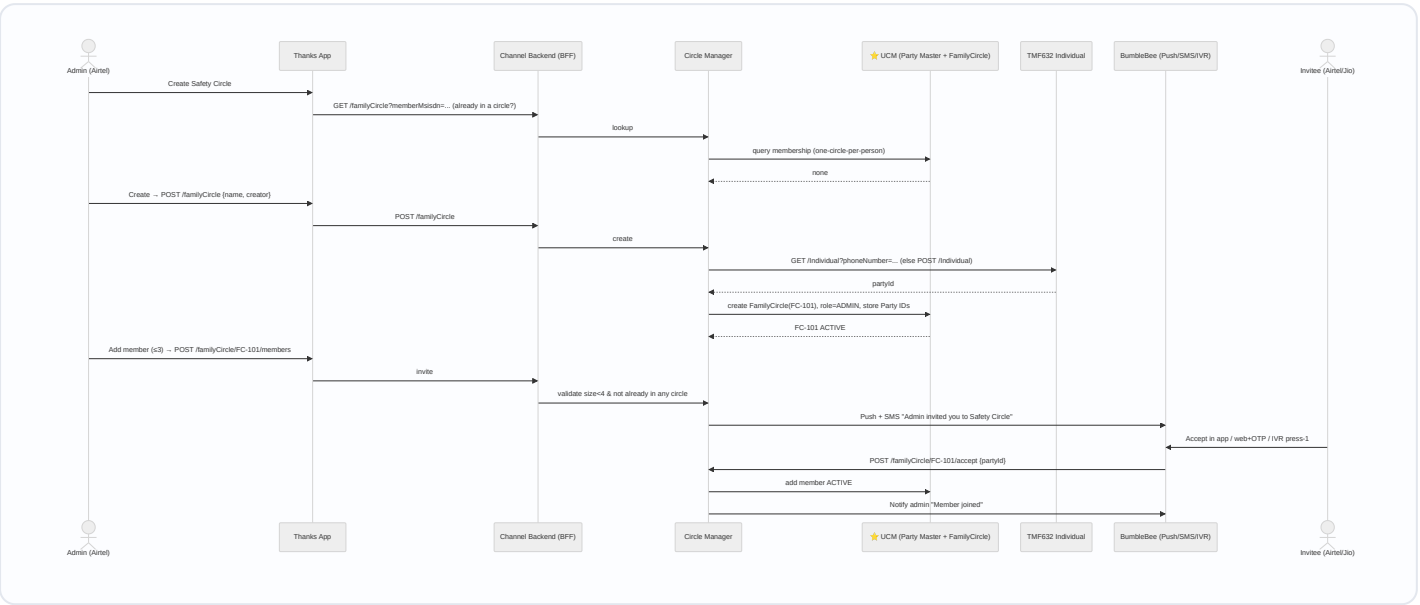
4.1 Context — everything around the Channel Backend (BFF)



4.2 Channel Backend (BFF) in focus — Device Safety toggle (EA sequence)



4.3 User/Party-profile backend in focus — Create circle + invite (UCM)



5. APIs by system (the core table)

5.1 Channel Backend (BFF) ⭐

Method	API	Purpose	Downstream	Source
GET	`/family-safety/home`	Aggregate landing: circle + members + card status	Circle, Telemetry, Network, Safety	`[INFERRED]` (Figma LP-35)
GET	`/family-safety/members/{id}`	Member detail aggregate	Circle, Telemetry, Safety	`[INFERRED]` (LP-38/39)
GET	`/family-safety/protection-status`	Device/Network Safety toggle states	Subscription Mgr	(EA)
POST	`/family-safety/device-protection`	Enable/disable Device Safety (config)	Subscription → Whalebone	(EA)
POST	`/family-safety/network-protection`	Enable/disable Network Safety (config)	Subscription → Network	(EA)
POST	`/family-safety/purchase` / `/claim`	Buy / claim pack	COM (TMF622)	(EA/BRD)
*	*pass-through* for circle/safety/trip	Thin proxy w/ auth + payload shaping	respective services	`[INFERRED]`

5.2 Circle Manager (on top of UCM) ⭐

Method	API	Purpose	Source
GET	`/familyCircle?memberMsisdn=`	Eligibility / already-in-circle	(EA)
POST	`/familyCircle`	Create circle	(EA) LP-04
GET	`/familyCircle/{id}`	Circle summary	(EA) LP-35
PATCH	`/familyCircle/{id}`	Name/avatar/settings	`[INFERRED]` LP-18
POST	`/familyCircle/{id}/members`	Invite (≤3; validate size<4)	(EA) SC1–4
GET	`/familyCircle/{id}/invites`	Pending invites	`[INFERRED]` US7
POST	`/invites/{id}/remind`	Remind (does **not** restart 2-day window)	`[INFERRED]` LP-09
POST	`/invites/{id}/resend`	Resend expired	`[INFERRED]` LP-17
GET	`/invites/{token}`	Invite preview (app/web)	(EA) LP-11..14
POST	`/familyCircle/{id}/accept`	Accept in app	(EA) LP-12
POST	`/invites/{token}/accept-web`	Accept after OTP	(EA) LP-13
POST	`/invites/{token}/accept-ivr`	Press-1 join	(EA) LP-14
POST	`/familyCircle/{id}/switch`	Leave other circle → join	`[INFERRED]` LP-16
GET	`/familyCircle/{id}/members`	Members + roles + relationship	(EA) SC2/SC5
PATCH	`/members/{memberId}`	Relationship/role/permissions	(EA) SC5/SC7
POST	`/familyCircle/{id}/members/{partyId}/leave`	Leave / leave-request	(EA) SC8
POST	`/leaveRequests/{id}/approve`	Admin approve/reject leave	(EA)
POST	`/familyCircle/{id}/adminExit`	Admin exit → grace period	(EA)
DELETE	`/members/{memberId}`	Admin remove	(EA) REMOVED
POST	`/members/{memberId}/reverify`	MSISDN/SIM change / 3-month non-Airtel re-verify	(EA) SC9
GET	`/contacts/airtel-family`	Airtel family first, then contacts	`[INFERRED]` LP-05·3

Events: `InviteCreated`, `InviteAccepted`, `MemberLeft`, `CirclePendingClosure`, `CircleTerminated`. **UCM data:** `FamilyCircle{circle_id, name, creator_id, status}` + member Party IDs + roles + per-member feature characteristics (new entity — **not** Customer/TMF629, not `Individual.relatedParty`).

5.3 Safety Manager

Method	API	Purpose	Source
POST	`/safety-events`	Create (`CHECK_IN` / `SOS` / `GEOFENCE` / `INACTIVITY` / `FRAUD` / `FALL`)	(EA) CI1/SO1
GET	`/safety-events/{id}`	Detail + state machine	(EA) CI5
GET	`/circles/{id}/safety-events`	History (30-day)	(EA) LP-40
PATCH	`/safety-events/{id}`	State transitions	(EA) CI3
POST	`/safety-events/{id}/responses`	Receiver / IVR response	(EA) CI2
GET	`/safety-events/{id}/responses`	Response list	(EA) CI5
POST	`/check-ins/schedules` + `GET` / `PATCH`	Scheduled check-in (repeat/pause/resume)	`[INFERRED]` LP-67/69
POST	`/sos/{id}/cancel`	Cancel + reason	(EA) SO5
PUT	`/circles/{id}/sos-priority`	SOS priority recipients	`[INFERRED]`
POST	`/sos/{id}/emergency-trip`	Start Emergency Trip	(EA)

Check-in state: `SENT→DELIVERED→RESPONDED | NO_RESPONSE→ESCALATING→ESCALATED` **SOS state:** `SENT→FAMILY_NOTIFIED→ACTIVE→RESOLVED | CANCELLED` **Escalation (CI4):** t+30 push → t+45 sticky → t+60 IVR → alert circle (thresholds via Admin Console).

5.4 Trip Manager + Location Hub (SafeTrip)

Method	API	System	Source
POST	`/trips` · `/trips/{id}/start` · `/stop` · `GET /trips/{id}` · `GET /trips/active` · `PATCH /trips/{id}`	Trip Manager	(EA)
POST	`/location/v1/tracking-sessions`	Location Hub — start session	(EA)
POST	`/location/v1/tracking-sessions/{TS}/locations`	Location Hub — ingest pings	(EA)
GET	`/location/stream?trackingSessionId={TS}`	Location Hub — **SSE** to viewers	(EA)
GET	`/location/v1/tracking-sessions/{TS}/latest`	Latest before SSE subscribe	`[INFERRED]`

Rules: dynamic ping interval (active vs idle), GPS else cell-tower via Network API, do **not** persist cell-only tracks, close stream on trip end.

5.5 Network API + Device Telemetry

Capability	Proposed API	Source
Reachability / network state	`GET /network/v1/msisdn/{msisdn}/reachability`	`[INFERRED]` UR1
Network event stream	`GET /network/v1/stream` (SSE)	`[INFERRED]`
Fraud signal feed	consume "Airtel Safe signal feed"	(BRD) FR1
Device heartbeat / status	`POST /telemetry/v1/devices/{id}/heartbeat` , `GET /telemetry/v1/members/{id}/status`	`[INFERRED]` LP-36

5.6 Commercial plane (COM / Subscription / Whalebone)

System	API / interface	Purpose
COM	TMF622 Product Order	All LOB orders for Family Safety PO
EPC + Promotion	Product/Service Spec (`partner_sku_id`), hidden free-sample PO	Catalog + 3-month postpaid trial
Subscription Mgr (ENDMA)	Create/update subscription with **Device Safety + Network Safety** services	Pack lifecycle, toggles
Whalebone Partner	allocate / map / verify / suspend / delete license; threat sync	`LicenseID → CustomerID → MSISDN → DeviceID → Subscription → Validity`
Whalebone SDK	on-device module inside Thanks App	malware/APK/URL/content/parental

5.7 Places / Geofence (Reachability + Parental) — ownership decision needed

Recommended: **Places owned by Circle Manager**, breach → Safety Manager `GEOFENCE` . `GET/POST /members/{id}/places` , `PUT /places/{id}/alerts` `[INFERRED]` .

---

6. Effort estimate (by API count + complexity)

Assumptions: one backend engineer ≈ 5 productive days/week; estimates are **backend build + integration + unit/integration test**, excluding app FE, partner-side work, SIT/UAT, and infra hardening. Complexity: **L** = CRUD/thin, **M** = state machine / multi-system, **H** = realtime / partner / data-model change.

#	System	APIs (new/changed)	Cx	Eng effort (person-weeks)	Key cost driver
1	**Channel Backend (BFF)** ⭐	~10 (aggregation + toggles + proxy)	M	**6–8**	AuthZ, home/member aggregation fan-out, error/degradation handling
2	**UCM — FamilyCircle entity + Party Master** ⭐	new entity + reads/writes	H	**8–12**	**New data model** (avoid Customer/Party collisions), migrations, 30-mesh problem
3	**Circle Manager**	~22	M–H	**10–14**	Invite lifecycle, consent, 3-channel accept, grace period, roles
4	**Safety Manager**	~13	M–H	**8–12**	Two state machines, responses, schedules, history/retention
5	**CLM escalation + IVR callback**	journey + callback	M	**3–5**	Staged timers, clickstream dependency, IVR integration
6	**Trip Manager**	~6	M	**4–6**	Trip lifecycle, viewer validation
7	**Location Hub**	~5 + pipeline	H	**8–12**	Ingest + Kafka + Redis + **SSE** fan-out + battery-aware intervals
8	**Network API**	~2 (+feed)	M	**3–5**	Reachability semantics, SSE vs poll, fraud feed
9	**Device Telemetry**	~2	M	**3–4**	Permission-gated capture, OS differences
10	**Push/SMS/BumbleBee + OTP**	integration	L–M	**3–5**	High-priority/sticky push, web-invite OTP
11	**COM + Subscription Mgr + EPC/Promotion**	TMF622 + service model	H	**10–14**	Order orchestration, trial PO, two-service subscription
12	**Whalebone integration + License Mgmt**	partner APIs + LMS	H	**8–12**	License inventory/rotation/mapping, suspend/delete, device re-pair
13	**Places / Geofence**	~3	M	**3–5**	Geofence eval + event routing
14	**Advisor view + Analytics/instrumentation**	read APIs + events	L–M	**3–5**	Privacy-scoped read, ~30 instrumentation events
—	**Cross-cutting** (consent model, privacy guardrails, security review, env, NFR)	—	H	**6–10**	Consent/permission split, "degrade gracefully", audit

Indicative backend total: ~95–130 person-weeks (≈ **2–3 pods over ~5–6 months** for MVP → full suite). **MVP (P0–P1: circle + check-in + SOS + push):** ~30–40 person-weeks. Add App FE (parallel, similar order), partner lead time, SIT/UAT separately.

Phased delivery (aligned to BRD roadmap)

Phase	Scope	Systems online
P0	Circle create/invite/accept (app+SMS+IVR), members, roles	BFF, Circle Mgr, UCM, BumbleBee, OTP
P1	Check-in + SOS + history + push escalation	+ Safety Mgr, CLM, IVR
P2	Telemetry + reachability on member cards	+ Network, Telemetry
P3	SafeTrip + Location Hub SSE	+ Trip Mgr, Location Hub, Redis/Kafka
P4	Commercial pack + Whalebone Device/Network Safety	+ COM, Subscription, EPC, Whalebone
P5	Parental Controls + Places/geofence + scheduled check-ins	+ Places, Whalebone parental
P6	Fraud / routine / unreachability intelligence polish	Network fraud, CLM tuning

---

7. Challenges / risks

#	Challenge	Impact	Mitigation
1	**New UCM FamilyCircle entity** — Customer (TMF629) and `Individual.relatedParty` both break (non-Airtel members, circle-level attrs, N×(N-1) mesh writes on join/leave)	Data-model rework if wrong	Build the dedicated FamilyCircle entity now (per EA); never model members as Customers
2	**Non-Airtel members (Jio/Vi)**	Reachability/network/fraud signals unavailable → degraded features	App-based check-in/SOS only; disclose limits before invite; 3-month re-verify
3	**Invite cap conflict** — Figma says 3, EA says max 4 (1 admin + 3) / older text 6	Data model + UX	**Resolve with Product** (plan uses 1 admin + 3 = 4)
4	**Realtime battery drain** (Location SSE)	UX/battery complaints	Dynamic active/idle ping; cell-tower fallback; don't persist cell-only tracks
5	**"Unreachable" ≠ "unsafe"** copy risk	Trust / false alarm	Strict copy review (UR2); explainable alerts; graceful degradation ("Unable to check status")
6	**Consent vs OS permission split**	Legal / privacy	Consent once at accept; permissions per-feature, independently revocable; immediate revocation
7	**Whalebone license lifecycle** — annual pool rotated into monthly/6-month packs, X-day retention, device re-pair, suspend/delete real-time?	Partner dependency, billing disputes	Confirm verify/disable SLA with partner; license inventory + mapping + reconciliation; define X days
8	**Toggle vs order** — Device/Network Safety enable/disable is **config on existing service, not a product order**	Billing correctness	Route toggles through Subscription Mgr characteristic change, not COM
9	**Admin exit / circle grace period**	Continuity	7-day grace, any eligible member can take over; revoke shared pack on exit
10	**Escalation timing not frozen** (5/10/15 min? CI4 says 30/45/60)	Behaviour inconsistency	Freeze thresholds with CFT; keep configurable via Admin Console
11	**iOS capability gaps** (Safari extension; limited background)	Feature parity	Disclose iOS limits in product (NFR-09)
12	**Parental Controls offline enforcement + uninstall/accessibility block**	Enforcement reliability	Local rules + sync on reconnect; block uninstall / alert parent (Android); declare device/OS-dependent
13	**Advisor privacy**	Compliance	Read-only, consent-scoped, no raw location/routine; OTP for assisted changes

#	Challenge	Impact	Mitigation
14	**Many systems (~24) + external partner**	Integration/SIT complexity	Phase delivery P0–P6; contract-first OpenAPI; BFF isolates app from churn

8. Open architecture decisions (owners to confirm)

#	Decision	Owner
1	Single Family Safety **BFF** vs app→each service	Architect (confirmed: BFF)
2	**Places** ownership (Circle vs Safety vs Location)	Architect
3	Invite max (3 vs 4 vs 6)	Product
4	Whalebone inside Thanks App module vs separate Safety App (EA: module; SDK BRD: separate app)	Product + Architect
5	Network SSE vs poll	Network + App
6	License X-day retention after expiry	Product + Partner
7	Safety Event delete policy / 30-day retention	Product
8	Escalation thresholds (30/45/60 vs 5/10/15)	CFT
9	Home Wi-Fi / IPTV / STB signal owners (routine break)	Homes + Network

9. Why the Customer / Party entity fails (data-model note)

- **Customer (TMF629)** = commercial billing relationship → modelling non-Airtel members as customers creates fake/shadow BSS accounts, pollutes metrics and CRM. ❌
- `Individual.relatedParty` = person-to-person pointer → can't hold circle-level attributes (name, avatar, rules) and forces N×(N-1) mutual-link writes (6 members = 30 links) on every join/leave. ❌
- ✅ **Solution:** a dedicated **UCM FamilyCircle entity** holding circle metadata, member Party IDs, roles and per-member feature characteristics — this is the "user/party-profile backend" for Family Safety. Commercial identity stays in Siebel/Customer; family hierarchy comes from Optimus; only the **admin line** carries the paid subscription.

10. Figma `39:4390` API surface — traceability, complexity & story points

This section takes the **exact API surface needed to build the Exhaustive-Cases canvas** and cross-checks every endpoint against the BRD and the EA HLD/LLD docs.

Columns

- **Backed by** — `EA` (named in an EA HLD), `BRD` (stated in a BRD), `Figma-gap` (needed by the design but **not** in any BRD/EA doc → raise as a BRD change request).
- **Owning system** — service that implements the business logic.
- **BFF integrates with** — if the Channel Backend (BFF) must build new orchestration for this API, the downstream systems it has to call.
- **Cx** — complexity: **L** (thin CRUD), **M** (state / multi-system), **H** (realtime / partner / new data-model).
- **SP (≈d)** — story points (Fibonacci) and the approximate **dev-days for 1 backend engineer** (1→½d, 2→1d, 3→2d, 5→3.5d, 8→6d).
- **Challenge** — the main risk for that endpoint.

> **SP→day guide:** 1=½d · 2=1d · 3=2d · 5=3.5d · 8=6d · 13=12d. SP here cover **backend build + integration + tests** only (not app FE, not partner-side work). Platform builds (realtime/SSE, scheduler, COM/subscription, Whalebone license) are tracked in §6.

Auth / session

API	Backed by	Owning system	BFF integrates with	Cx	SP (≈d)	Challenge
`GET /me`	BRD (shell)	BFF + UCM/Customer	UCM, Customer/Siebel	L	2 (1d)	Identity resolution across app-shell + circle
`GET /eligibility/safety-circle`	EA (design assumptions)	Circle Mgr + Subscription	CM, Subscription, Siebel, Audience Mgr	M	3 (2d)	Eligibility rules not finalised; needs active-pack check

1. Getting started (11 SP ≈ 7d)

API	Backed by	Owning system	BFF integrates with	Cx	SP (≈d)	Challenge
`POST /circles`	EA (`POST /familyCircle`)	Circle Mgr → UCM	CM	M	5 (3.5d)	New FamilyCircle entity; one-circle-per-person rule
`GET /circles/{id}`	EA	Circle Mgr	CM	L	2 (1d)	—
`GET /onboarding/permissions`	**Figma-gap**	BFF / CMS	CMS	L	1 (½d)	Content mgmt; copy localisation
`POST /device/permissions`	BRD (permissions)	Device Telemetry	TEL	M	3 (2d)	OS permission capture Android vs iOS

2. Adding and joining (33 SP ≈ 21.5d)

API	Backed by	Owning system	BFF integrates with	Cx	SP (≈d)	Challenge
`GET /contacts/airtel-family`	EA (design)	Circle Mgr → Optimus	CM, Optimus/Siebel	M	5 (3.5d)	Optimus family-mode fetch; PII/consent
`POST /circles/{id}/invites`	EA	Circle Mgr + BumbleBee	CM, BB(Push/SMS)	M	5 (3.5d)	Cap enforcement; dual-channel dispatch
`POST /invites/{id}/remind`	EA	Circle Mgr + BB	CM, BB	L	2 (1d)	Must **not** restart 2-day window
`POST /invites/{id}/resend`	EA (resend)	Circle Mgr + BB	CM, BB	L	2 (1d)	New token, invalidate old
`GET /circles/{id}/invites`	Figma-gap	Circle Mgr	CM	L	2 (1d)	—
`GET /invites/{token}`	EA	Circle Mgr	CM	L	2 (1d)	App + web + SMS preview parity
`POST /invites/{token}/accept`	EA	Circle Mgr → UCM	CM	M	3 (2d)	Consent capture; membership write
`POST /invites/{token}/accept-ivr`	EA (LP-14)	IVR/BumbleBee → Circle Mgr	CM, IVR	H	5 (3.5d)	IVR integration; token↔MSISDN binding
`POST /invites/{token}/decline`	EA (lifecycle)	Circle Mgr	CM	L	1 (½d)	—
`POST /circles/switch`	Figma-gap (LP-16)	Circle Mgr → UCM	CM	M	3 (2d)	Atomic leave-old + join-new
`GET /members/{msisdn}/availability`	EA (inferred)	Circle Mgr / UCM	CM	M	3 (2d)	Already-in / already-invited lookup

> Note: EA also defines `POST /invites/{token}/accept-web` (OTP) — add to the Figma set for the no-app web-join (LP-13).

3. Managing the circle (21 SP ≈ 14d)

API	Backed by	Owning system	BFF integrates with	Cx	SP (≈d)	Challenge
`GET /circles/{id}/members`	EA	Circle Mgr	CM	L	2 (1d)	—
`PATCH …/members/{id}/role`	EA (SC5/SC7)	Circle Mgr + OTP	CM, OTP/Bouncer	M	3 (2d)	OTP/biometric auth for role change
`DELETE …/members/{id}`	EA (REMOVED)	Circle Mgr → UCM + BB	CM, BB, Subscription	M	3 (2d)	De-provision sharing immediately
`POST /circles/{id}/leave`	EA (SC8)	Circle Mgr + Subscription + BB	CM, Subscription, BB	M	5 (3.5d)	Parental-lock, admin approval, grace period
`GET/PATCH /circles/{id}/settings`	Figma-gap (LP-18)	Circle Mgr	CM	L	3 (2d)	—
`POST /circles/{id}/protection/stop`	EA (SC9/MC-04)	Circle Mgr + Subscription + Network	CM, Subscription, Network	H	5 (3.5d)	Port-out / SIM-swap detection; re-verify

4. Home and member pages (23 SP ≈ 15.5d)

API	Backed by	Owning system	BFF integrates with	Cx	SP (≈d)	Challenge
`GET /circles/{id}/home`	EA (DB1)	**BFF aggregate**	CM, Telemetry, Network, Safety Mgr	H	8 (6d)	Multi-system fan-out, partial-failure, latency
`GET /circles/{id}/members/{id}`	EA (DB3)	**BFF aggregate**	CM, Telemetry, Safety Mgr	M	5 (3.5d)	Compose profile + status + history
`GET /members/{id}/device-status`	BRD (§8 signals)	Telemetry + Network	TEL, Network	M	3 (2d)	Permission-gated; non-Airtel gaps
`GET /circles/{id}/history`	EA (CI5)	Safety Mgr	SM	M	3 (2d)	30-day retention policy
`GET /alerts/{alertId}`	EA	Safety Mgr	SM	L	2 (1d)	—
`GET /me/sharing`	BRD (DB4 privacy)	Circle Mgr / Telemetry	CM, TEL	L	2 (1d)	"What the circle sees" transparency

5. Reachability — places & alerts (24 SP ≈ 16.5d)

API	Backed by	Owning system	BFF integrates with	Cx	SP (≈d)	Challenge
`GET/POST /members/{id}/places`	Figma-gap (BRD has reachability, not places CRUD)	Places (Circle Mgr) + Maps	CM/Places, Maps	M	5 (3.5d)	Geocoding; **Places ownership undecided (§8)**
`PATCH/DELETE /places/{placeId}`	Figma-gap	Places	CM/Places	L	2 (1d)	—
`PUT /places/{placeId}/alerts`	Figma-gap	Places → Safety Mgr (GEOFENCE)	CM/Places, SM	M	3 (2d)	Geofence event routing
`POST /location/ping`	EA (Safety Trip)	Location Hub	LH	M	3 (2d)	Battery; dynamic interval
`GET /members/{id}/location`	EA + BRD (UR)	Location Hub + Network (cell)	LH, Network	M	3 (2d)	Consent; location-off state
`WS /realtime/location`	EA (SSE)	Location Hub	LH (Kafka/Redis/SSE)	H	8 (6d)	Realtime infra + scale + battery

6. SOS (20 SP ≈ 13d)

API	Backed by	Owning system	BFF integrates with	Cx	SP (≈d)	Challenge
`POST /sos`	EA (SO1)	Safety Mgr	SM	M	5 (3.5d)	Countdown, active-state machine
`POST /sos/{id}/end`	EA (SO3/SO5)	Safety Mgr	SM	L	2 (1d)	"I'm safe" + reason
`GET /sos/active`	EA	Safety Mgr	SM	L	2 (1d)	—
`PUT /circles/{id}/sos-priority`	Figma-gap	Safety Mgr	SM	L	3 (2d)	Priority-recipient model
`POST /sos/{id}/location`	EA (SO2)	Safety Mgr + Location + Network	SM, LH, Network	M	3 (2d)	Location-off disclosure
`POST /sos/{id}/notify`	EA (SO2)	Safety Mgr + BumbleBee	SM, BB(Push/SMS/IVR)	H	5 (3.5d)	Critical push + SMS + call reliability

7. Fraud alerts (15 SP ≈ 9.5d)

API	Backed by	Owning system	BFF integrates with	Cx	SP (≈d)	Challenge
`GET /fraud/alerts`	BRD (FR) + EA	Safety Mgr (FRAUD)	SM	M	3 (2d)	—
`GET /fraud/alerts/{id}`	BRD	Safety Mgr	SM	L	2 (1d)	—
`PATCH /fraud/alerts/{id}/ack`	BRD	Safety Mgr	SM	L	2 (1d)	—
`PUT /members/{id}/fraud-settings`	BRD (FR2)	Circle Mgr / Safety Mgr	CM/SM	L	3 (2d)	Opt-in family fraud protection
Ingest webhook (Device Safety / Network)	BRD (FR1 feed)	Network + Whalebone → Safety Mgr	Network, Whalebone, SM	H	5 (3.5d)	Partner/network signal ingest, dedup

8. Check-ins (25 SP ≈ 16d)

API	Backed by	Owning system	BFF integrates with	Cx	SP (≈d)	Challenge
`POST /check-ins`	BRD/EA (CI1)	Safety Mgr	SM	M	3 (2d)	Single-open-check-in rule
`POST /check-ins/{id}/reply`	EA (CI2)	Safety Mgr (+IVR)	SM, IVR	M	3 (2d)	Preset + custom + IVR-equivalent reply
`GET /members/{id}/check-ins`	EA (CI3)	Safety Mgr	SM	L	2 (1d)	—
`POST /check-ins/schedules`	**Figma-gap** (BRD is manual-first)	Safety Mgr + Scheduler	SM, Scheduler	M	5 (3.5d)	Scheduler infra; repeat/time model
`PATCH /check-ins/schedules/{id}`	Figma-gap	Safety Mgr	SM	L	3 (2d)	Pause / resume
`DELETE /check-ins/schedules/{id}`	Figma-gap	Safety Mgr	SM	L	2 (1d)	—
`GET /check-ins/schedules`	Figma-gap	Safety Mgr	SM	L	2 (1d)	—
Scheduler job + push	Figma-gap	Scheduler + CLM + BumbleBee	Scheduler, CLM, BB	H	5 (3.5d)	Quiet notif + escalation + SMS/IVR fallback

9. Device Safety (25 SP ≈ 17d)

API	Backed by	Owning system	BFF integrates with	Cx	SP (≈d)	Challenge
`POST /device-safety/enable`	EA + SDK BRD	Subscription + Whalebone	Subscription, Whalebone	H	5 (3.5d)	License allocate; partner SLA
`GET /members/{id}/device-safety`	EA (LP-78)	Subscription / Whalebone	Subscription, Whalebone	M	3 (2d)	Status + what circle sees
`POST /device-safety/permissions`	BRD (SDK)	Whalebone SDK / Telemetry	TEL, Whalebone SDK	M	3 (2d)	Stepwise Android; iOS Safari limits
`POST /device-safety/scan`	SDK BRD (F1)	Whalebone SDK	Whalebone SDK	M	3 (2d)	First-scan auto-run
`POST /device-safety/events`	SDK BRD (F11)	Whalebone → ingest	Whalebone, SM/Dashboard	H	5 (3.5d)	Partner event sync; volume
`GET /device-safety/events`	SDK BRD (F6)	Safety Mgr / Dashboard store	SM	M	3 (2d)	Admin history
`POST /device-safety/invite-install`	EA (LP-81)	BumbleBee SMS + deep link	BB, Whalebone	M	3 (2d)	Install deep-link path

10. Parental Controls (34 SP ≈ 23.5d)

API	Backed by	Owning system	BFF integrates with	Cx	SP (≈d)	Challenge
`POST /parental/pair`	SDK BRD (F4)	Whalebone SDK + Circle Mgr	Whalebone, CM	H	5 (3.5d)	Pairing-code generation; child consent
`POST /parental/pair/confirm`	SDK BRD (F4)	Whalebone + Telemetry	Whalebone, TEL	M	3 (2d)	Code entry = child's yes; permissions
`GET/PUT /members/{id}/parental`	SDK BRD (F4)	Whalebone (rules sync)	Whalebone	H	8 (6d)	Screen-time/app/web rules + offline sync
`GET /members/{id}/parental/usage`	Figma-gap	Whalebone	Whalebone	M	5 (3.5d)	Usage by app/category/day aggregation
`GET /parental/requests`	Figma-gap	Safety Mgr / Circle Mgr	SM/CM	M	3 (2d)	—
`POST /parental/requests/{id}/decide`	Figma-gap	Safety Mgr + BumbleBee	SM, BB	M	5 (3.5d)	Multi-parent approve/decline; expiry
`POST /parental/disable`	SDK BRD + Figma	Whalebone + Circle Mgr	Whalebone, CM	M	3 (2d)	Uninstall/accessibility block + alert
`PUT /members/{id}/places` (child)	Figma-gap	Places (Circle Mgr)	CM/Places	L	2 (1d)	Reuse of Places service

Roll-up

Section	Story points	≈ dev-days (1 BE)	Figma-gaps (new scope vs BRD/EA)
Auth / session	5	3	0
1 Getting started	11	7	1 (`/onboarding/permissions`)
2 Adding & joining	33	21.5	2 (`/circles/{id}/invites` GET, `/circles/switch`)
3 Managing the circle	21	14	1 (`/settings`)
4 Home & member pages	23	15.5	0
5 Reachability	24	16.5	3 (places CRUD + alerts)
6 SOS	20	13	1 (`sos-priority`)
7 Fraud alerts	15	9.5	0
8 Check-ins	25	16	5 (all scheduling)
9 Device Safety	25	17	0
10 Parental Controls	34	23.5	3 (usage + requests)
TOTAL (API layer)	**≈236 SP**	**≈156 dev-days (~31 person-weeks)**	**16 Figma-gaps**

> **Reading this:** ~156 dev-days ≈ **31 person-weeks of backend API work** for the whole canvas. This is the **API-build layer only**. The platform builds behind it — realtime/SSE (Location Hub), scheduler, COM + Subscription Manager, Whalebone license management, CLM escalation — are the larger cost and are estimated in §6 (≈95–130 person-weeks full program). **16 of ~60 endpoints are "Figma-gaps"** (designed but not in any BRD/EA doc) — mostly **Places CRUD** and **scheduled check-ins** — and must be ratified by Product as scope before committing a sprint plan.

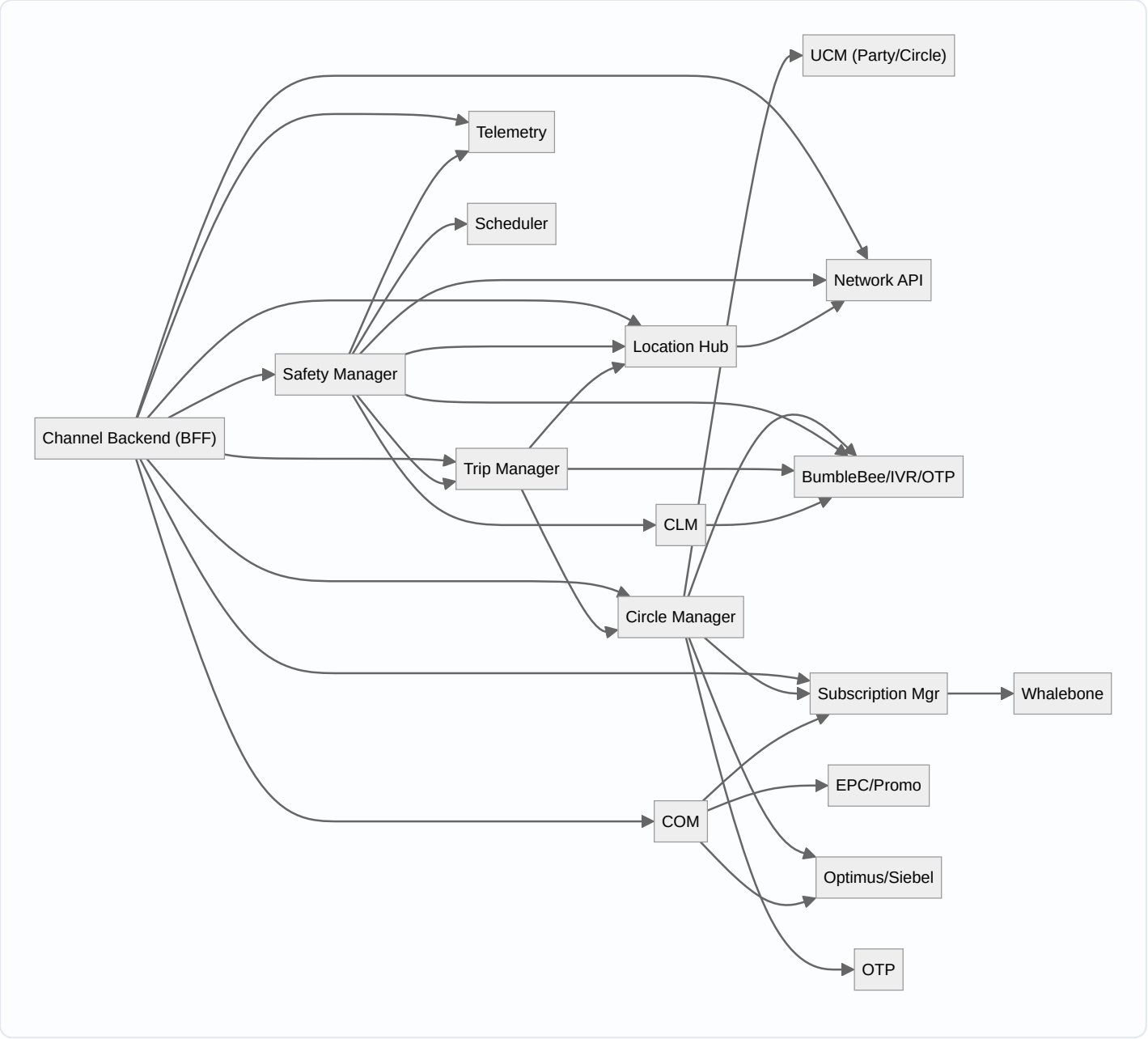
---

11. System integration & dependency map

Which systems a given system **calls** (→ depends on), and how tightly coupled.

System	Depends on (calls)	Coupling	Note
Channel Backend (BFF)	Circle Mgr, Safety Mgr, Trip Mgr, Location Hub, Network, Telemetry, COM, Subscription	**Very high**	Single front door; a change in any downstream can ripple to BFF payloads
Circle Manager	UCM, TMF632 Individual, Optimus/Siebel, OTP/Bouncer, BumbleBee, Subscription	**High**	Core of circle + membership; most-integrated service
UCM (FamilyCircle + Party)	— (system of record)	Low (inbound)	Everyone reads/writes members here; schema stability critical
Safety Manager	BumbleBee, CLM, Trip Mgr, Location Hub, Network, Telemetry, Scheduler	**High**	Hub of check-in/SOS/fraud/geofence events
Trip Manager	Circle Mgr, Location Hub, BumbleBee	Medium	Validates members, owns trip lifecycle
Location Hub	Network (cell fallback), Kafka/Redis/History	Medium	Realtime pipeline; internal infra heavy
Network API	— (signal source)	Low (inbound)	Reachability + fraud feed provider
Device Telemetry	— (signal source)	Low (inbound)	Battery/signal/permission provider
CLM	Clickstream, BumbleBee, IVR, Safety Mgr	Medium	Escalation orchestration
COM (Service Order Mgr)	EPC, Promotion, Subscription, Siebel/ATOM/FX	**High**	TMF622 order spine
Subscription Mgr (ENDMA)	COM, Whalebone	**High**	Two-service subscription; partner provisioning
Whalebone (partner)	— (external)	External	License + threat sync; SLA-bound
BumbleBee / IVR / OTP	— (platforms)	Low (inbound)	Notification + auth utilities

Tightest coupling to watch: `BFF ↔ Circle Mgr ↔ UCM` (every journey) and `COM ↔ Subscription ↔ Whalebone` (every paid activation). Contract-first OpenAPI + the BFF indirection layer are the two levers that keep this from becoming brittle.



12. Channel Backend (BFF) — new APIs and their required integrations

For every API where the **BFF must build new orchestration** (not a thin pass-through), the systems it has to integrate with to fulfil the requirement, plus the specific risk.

New BFF API	Must integrate with	Why (orchestration)	Cx	SP (≈d)	Challenge
`GET /me`	UCM, Customer/Siebel	Merge app-shell identity + circle membership	L	2 (1d)	Stitching identity across shell + circle sources
`GET /eligibility/safety-circle`	Circle Mgr, Subscription, Siebel, Audience Mgr	Check one-circle rule + active pack + trial eligibility	M	3 (2d)	Eligibility rules unowned; prepaid vs postpaid
`GET /circles/{id}/home`	Circle Mgr, Telemetry, Network, Safety Mgr	Build landing: members + device cards + alerts + quick actions	H	8 (6d)	Partial failures must degrade gracefully; latency budget; caching
`GET /circles/{id}/members/{id}`	Circle Mgr, Telemetry, Safety Mgr	Compose member page (profile + status + history)	M	5 (3.5d)	Fan-out + consistency across 3 sources
`GET /members/{id}/device-status`	Telemetry, Network	Merge battery/signal/mode/reachability	M	3 (2d)	Non-Airtel members → missing signals; "unknown ≠ off"
`GET/POST protection toggles` (`/family-safety/device-protection` , `/network-protection` , `/protection-status`)	Subscription Mgr (→ Whalebone / Network)	Toggle = **config on existing service, not an order**	M	5 (3.5d)	Must not create a COM product order; verify ACTIVE/TRIAL
`POST /family-safety/purchase` / `/claim`	COM (TMF622), EPC/Promotion, Subscription	Place order / claim free-sample PO	H	8 (6d)	TMF622 orchestration; hidden trial PO; eligibility
`POST /sos/{id}/notify` (if fanned via BFF)	Safety Mgr, BumbleBee	Critical fan-out push + SMS + call	H	5 (3.5d)	Delivery guarantees; prefer Safety Mgr to own
Pass-through (circle/safety/trip CRUD)	respective owning service	Auth + payload shaping only	L	1–2 ea	Keep thin; don't leak business logic into BFF

Net-new BFF orchestration effort ≈ 39 SP (~26 dev-days), dominated by the two aggregators (`/home` , member page) and the commercial `purchase/claim` + protection toggles. Everything else the BFF exposes for this canvas should be a **thin authenticated proxy** to the owning service.

This is an integration HLD/LLD planning doc derived from the 5 uploaded PDFs + Figma `39:4390`. It is not a substitute for the EA LLD packages marked "LLD — Implementation team to share." Finalise §5/§10 as OpenAPI per service and resolve §8 before build. Story points are planning-grade estimates for backend build+integration+test; confirm with the delivery pod during sprint-0.